

DB Design

Collections:

User

Item

Category

Exchange Request

Exchange Room

Message

Notification

Review

Relationships:

USER —————> ITEMS <----- CATEGORY

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EXCHANGE REQUEST

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▼

EXCHANGE ROOM

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MESSAGES

USER —————> NOTIFICATIONS

USER —————> REVIEWS

Collection 1 — Users

Field	Type	Required	Description
_id	ObjectId	✓	MongoDB ID
fullName	String	✓	User's name
email	String	✓	Login email
password	String	Optional	Email login
googleId	String	Optional	Google OAuth ID
profilePicture	String	✗	Avatar
phone	String	Optional	Contact number
bio	String	✗	User bio
location	String	✓	City/State
totalCompletedExchanges	Number	✓	Number of successfully completed exchanges
averageRating	Number	✓	Average rating received
exchangeSuccessRate	Number	✓	Percentage of accepted exchanges successfully completed
role	Enum	✓	USER / ADMIN
status	Enum	✓	ACTIVE / SUSPENDED
createdAt	Date	✓	Account creation date
updatedAt	Date	✓	Last updated date

Collection 2 — Categories

Field	Type	Required	Description
_id	ObjectId		MongoDB document ID
name	String		Name of the parent category (e.g., Electronics, Books, Furniture)
subcategories	Array		List of predefined subcategories belonging to the category (e.g., Laptop, Camera, Fiction)
icon	String		Icon representing the category
description	String		Brief description of the category
isActive	Boolean		Indicates whether the category is currently available for use
createdAt	Date		Category creation date
updatedAt	Date		Last updated date

Collection 3 — Items

Field	Type	Required	Description
_id	ObjectId	✓	MongoDB document ID
ownerId	ObjectId (User)	✓	References the owner in the Users collection
title	String	✓	Name of the item (e.g., "Canon EOS 200D")
description	String	✓	Detailed description of the item
categoryId	ObjectId	✓	References the Categories collection
condition	Enum	✓	NEW, LIKE_NEW, GOOD, FAIR, POOR
images	Array<String>	✓	URLs/paths of uploaded item images
videos	Array<String>	✗	URLs/paths of uploaded item videos
exchangePreferences	Array<Object>	✓	Ordered list of preferred exchange categories or subcategories arranged by priority
status	Enum	✓	AVAILABLE, PENDING, EXCHANGED, REMOVED
viewCount	Number	✗	Number of times the listing has been viewed
favoriteCount	Number	✗	Number of users who bookmarked the listing
exchangeRequestCount	Number	✗	Total exchange requests received
createdAt	Date	✓	Listing creation date
updatedAt	Date	✓	Last updated date

Collection 4 — Exchange Requests

Field	Type	Required	Description
_id	ObjectId	✓	MongoDB document ID
participants	Array<Object>	✓	Stores each participant, the item they are offering, and their acceptance status.
exchangeType	Enum	✓	DIRECT or THREE_WAY
cycleSize	Number	✓	Number of participants in the exchange (2 or 3 in Version 1.0)
status	Enum	✓	PENDING, ACCEPTED, REJECTED, CANCELLED, COMPLETED, EXPIRED
initiatedBy	ObjectId (User)	✓	User who initiated the exchange request
exchangeRoomId	ObjectId	✗	Reference to the Exchange Room created after all participants accept
createdAt	Date	✓	Exchange request creation date
updatedAt	Date	✓	Last updated date

Collection 5 — Exchange Rooms

Field	Type	Required	Description
_id	ObjectId		MongoDB document ID
exchangeRequestId	ObjectId (ExchangeRequest)		Reference to the accepted exchange request
participants	Array<ObjectId>		Users participating in the exchange
meeting	Object		Meeting details (location, date, time, status)
exchangeStatus	Enum		WAITING, MEETING_SCHEDULED, WAITING_CONFIRMATION, COMPLETED, DISPUTED
confirmationStatus	Array<Object>		Stores confirmation status of each participant
createdAt	Date		Room creation date
updatedAt	Date		Last updated date

Collection 6 — Messages

Field	Type	Required	Description
_id	ObjectId	✓	MongoDB document ID
exchangeRoomId	ObjectId (ExchangeRoom)	✓	Exchange Room to which the message belongs
senderId	ObjectId (User)	✓	User who sent the message
messageType	Enum	✓	TEXT, IMAGE, SYSTEM
message	String	✗	Text content (only for TEXT messages)
imageUrl	String	✗	Cloudinary image URL (only for IMAGE messages)
sentAt	Date	✓	Message timestamp

Collection 7 — Notifications

Field	Type	Required	Description
_id	ObjectId	✓	MongoDB document ID
recipientId	ObjectId (User)	✓	User receiving the notification
senderId	ObjectId (User)	✗	User who triggered the notification (null for system notifications)
notificationType	Enum	✓	Type of notification
title	String	✓	Short notification title
message	String	✓	Notification description
referenceId	ObjectId	✗	Related document (Exchange Request, Exchange Room, Item, Review, etc.)
referenceType	Enum	✗	EXCHANGE_REQUEST, EXCHANGE_ROOM, ITEM, REVIEW
isRead	Boolean	✓	Indicates whether the notification has been read
createdAt	Date	✓	Notification creation date

Collection 8 - Reviews

Field	Type	Required	Description
_id	ObjectId	✓	MongoDB ID
exchangeRoomId	ObjectId	✓	Completed exchange
reviewerId	ObjectId(User)	✓	Person giving review
revieweeId	ObjectId(User)	✓	Person receiving review
rating	Number	✓	1–5 stars
comment	String	✗	Optional review
createdAt	Date	✓	Review date

Final exchange flow for listing items:

User creates item listing



Matching Engine analyzes all listings



Finds valid 2-way or 3-way cycles



Shows exchange suggestions



User chooses one suggestion



Exchange Request created



Other participants notified



Everyone accepts?



No



Exchange ends



Yes



Exchange Room created



Chat & Meeting Planning



Exchange Completed



Ratings & Reviews

Final exchange flow for browsing:-

Browse Marketplace



Select Item



Choose one of your active listed items to offer



Send Direct Exchange Request



Exchange Request



Exchange Room



Complete Exchange



Ratings and Reviews